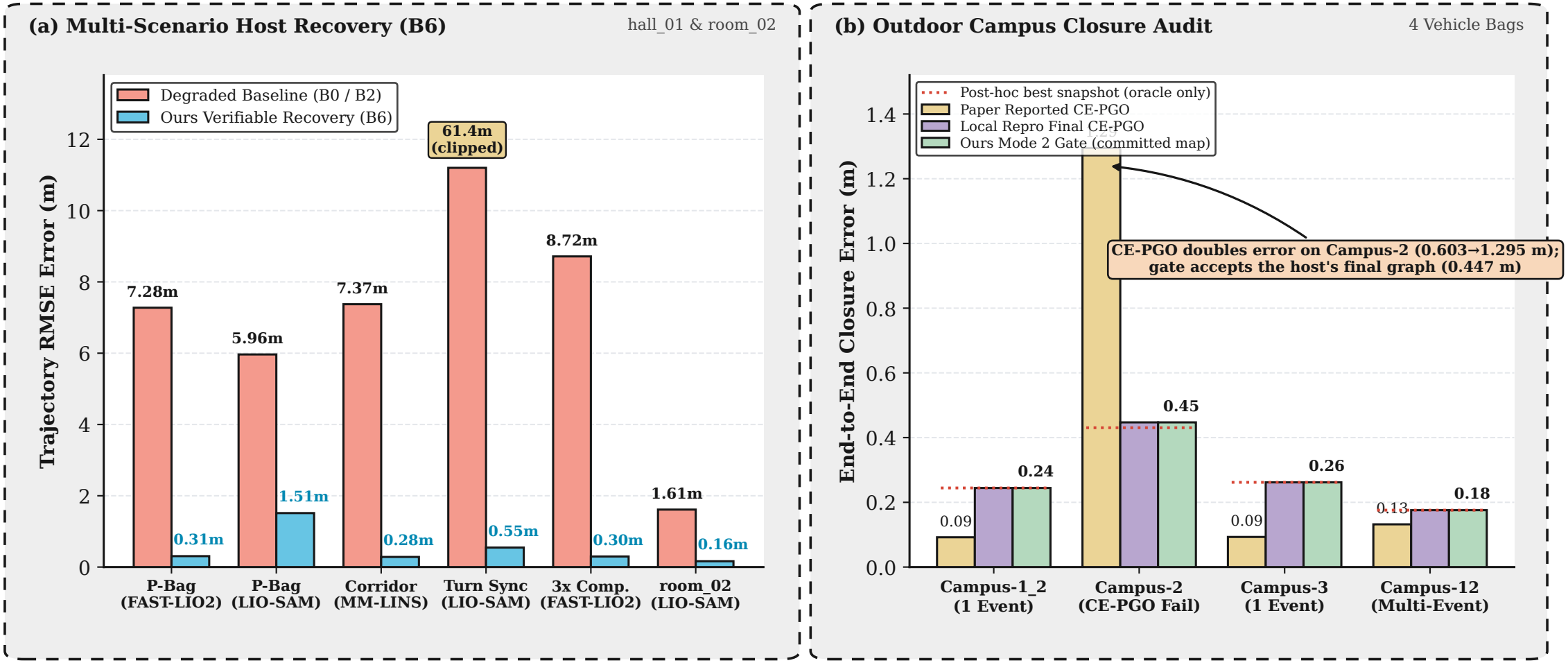
(a) Multi-Scenario Host Recovery (B6)
hall_01 & room_02
Degraded Baseline (B0 / B2)
Ours Verifiable Recovery (B6)
Trajectory RMSE Error (m)
0
2
4
6
8
10
12
7.28m
0.31m
5.96m
1.51m
7.37m
0.28m
61.4m
(clipped)
0.55m
8.72m
0.30m
1.61m
0.16m
P-Bag
(FAST-LIO2)
P-Bag
(LIO-SAM)
Corridor
(MM-LINS)
Turn Sync
(LIO-SAM)
3x Comp.
(FAST-LIO2)
room_02
(LIO-SAM)
(b) Outdoor Campus Closure Audit
4 Vehicle Bags
Post-hoc best snapshot (oracle only)
Paper Reported CE-PGO
Local Repro Final CE-PGO
Ours Mode 2 Gate (committed map)
End-to-End Closure Error (m)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
CE-PGO doubles error on Campus-2 (0.603→1.295 m);
gate accepts the host's final graph (0.447 m)
0.09
0.24
0.45
0.09
0.26
0.13
0.18
Campus-1_2
(1 Event)
Campus-2
(CE-PGO Fail)
Campus-3
(1 Event)
Campus-12
(Multi-Event)